

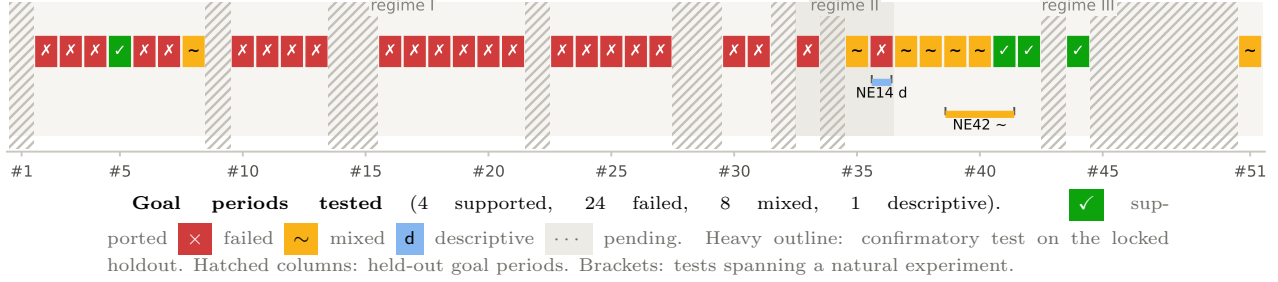
H67: A lagged criticality dial

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 2.5/5

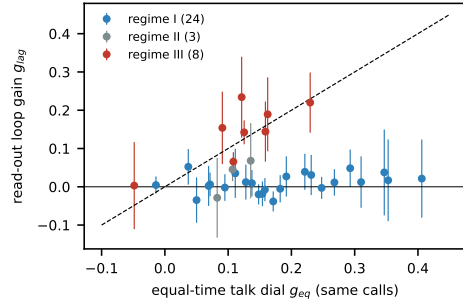
Bottom line. Read-out talk coupling is real but small and subcritical ($g_{lag} \approx 0.13$ in regime III, ≈ 0 elsewhere), it is carried by named messages, and the equal-time dial overstates it wherever fields drive talk.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H25's equal-time talk dial measures feedback within one minute. Coupling acts at the recipient's next model call, so HH262 asked whether a dial built on that read-out call shows larger gains that are still subcritical. The test covers 66 non-holdout units in 35 periods, with 1.47M receiving calls in all-present windows. The read-out loop gain g_{lag} counts the extra talk messages that one message causes through reading. Every unit is subcritical (all upper bounds < 0.40). The gain is 0.13 in regime III (median) and ≈ 0 in regimes I and II. It exceeds the equal-time dial in only 7/35 periods, so **the HH claim fails**. In regime III, messages that name the recipient carry about half of the gain. g_{lag} ranks periods like H42's read-out Hawkes fit ($\rho = 0.55$). The holdout is not run.



Observables. Each point is one goal period (random-effects pool of its units, 95% CI). The x-axis is H25's equal-time talk dial on the same calls, the y-axis the read-out loop gain. Regime-I periods sit on $g_{lag} \approx 0$ while g_{eq} reads 0.05–0.40: a fast shared field, not read-out coupling. Regime-III periods lie near the diagonal at 0.07–0.23.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: report g_{lag} , not the equal-time dial, as distance to a talk cascade. The equal-time dial counts shared fields as feedback, and in regime I that is all it sees. Here one talk message causes about 0.13 further messages through reading (amplification $\times 1.15$, $T/T_c \approx 8$). The gain stays flat as the swarm grows from 21 to 32 agents, because per-read coupling falls with headcount ($\rho = -0.59$). For an operator: naming carries the coupling. A named message raises the recipient's next-call talk probability by 0.08; an unnamed one by 0.004. No village configuration came near runaway, so this is a gauge, not a lever.

PHYSICAL MODEL

Linear response on the call clock (models 01, 09). Agent i 's call c is a talk call ($Y_{ic} = 1$) or not. R_{ic}^m counts peer messages read at c that were posted within the call's latency d_c before its start. P_{ic} counts messages posted within d_c after the start (in flight, unreadable). $a_{i,d,\kappa}$ is an agent $\times$ day $\times$ call-class field. $\bar{r}$ is the readers per message and $\bar{m}$ the messages per talk call.

$$P(Y_{ic}=1) = a_{i,d,\kappa} + \beta_m R_{ic}^m + \beta_o R_{ic}^o + \beta_P P_{ic} + \gamma \cdot E_{ic} + \rho Y_{i,c-1}$$

$$J_1^* = \beta_m - \beta_P, \quad g_{lag} = \bar{m} \bar{r} J_1^*, \quad T/T_c = 1/g_{lag}$$

$$g_{eq} = 1 - \left[\sum_t \sum_i X_{it}^2 \right] / \sum_t (\sum_i X_{it})^2$$

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RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1–S4 synthetic (real call grids)	bias -1 to -3% , null $CI \neq 0$ 7.5%; 3/4 pass equal-time dial not blind (S3)	
P1 subcritical everywhere	max upper bound 0.39; all units < 0.5	supported
P2 $g_{lag} > g_{eq}$ in $\geq 2/3$	7/35 periods; median ratio 0.09	failed
P3 rank-consistent with H42	$\rho = 0.55$ ($p = 0.0007$); ratio 2.0	supported
P3b consistent with H50	$\rho = 0.24$; H50 implies 0.24 vs 0.13	failed
P4 jump over placebo	$J_1^* > 0$ in 35% (III: 80%); shift size 13.6%	mixed
P5 named $\geq 5 \times$ un-named (III)	0.079 vs 0.004 ($\times 19$)	supported
P6 trim, exogenous robust	-10% , 0% (III medians)	supported
P7 J_1^* falls with N	$\rho = -0.59$; g_{lag} flat ($\rho = 0.24$)	supported
Natives NE14, NE42, G51	switch-on at II $\rightarrow$ III; #40 merge kills coupling; flat in N	d / mixed / mixed

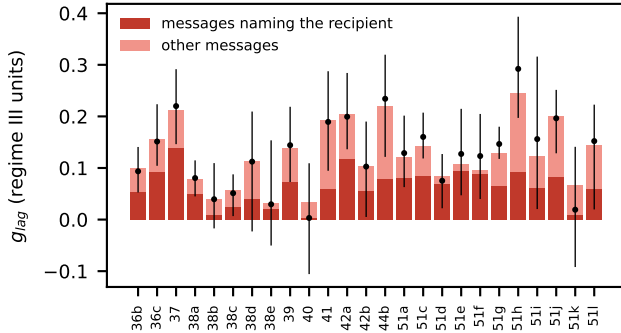
GOAL PERIODS IN DETAIL

G02 $\times$ failed
G03 $\times$ failed
G04 $\times$ failed
G05 $\checkmark$ supported
G06 $\times$ failed
G07 $\times$ failed
G08 $\sim$ mixed
G10 $\times$ failed
G11 $\times$ failed
G12 $\times$ failed
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Regime-III units ordered by headcount. Bars split g_{lag} into the part carried by messages that name the recipient (dark) and the rest (light). Points show g_{lag} with 95% CI. The named part is 0.03–0.14 in every unit. The unnamed part is small per read but is multiplied by 10–30 readers.

CAVEATS

The unnamed part (about 0.003 per read) sits at the resolution of the in-flight placebo. In-flight messages are decided closer to the call, which biases g by about ± 0.03 in synthetic burst worlds. A post hoc quiet-recipient subset lowers the regime-III median to 0.086. Regime-I talk happens at scheduled chat calls, so a hop-1 estimator may miss a later response there. Only direct offspring at hop 1 are placebo-corrected. Per-unit significance is mildly anti-conservative (shift size 13.6%). Predictions were written knowing H25, H42 and H50.

ROUND 2

Re-index regime I to its talk-capable calls. Confirm the named-message gain on the holdout (C3). Run H50's and H67's estimators on identical pairs to locate the factor-2 gap.

- **Essence.** a lag-aware, field-free loop gain that an operator can trust as "distance to runaway talk", on the clock the scaffold imposes.
- **H67-R1** Regime-I clock.
- **H67-R2** Named vs unnamed on the holdout.
- **H67-R3** Reconcile with H50.
- **H67-R4** Multi-generation gain.

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